SPE-226852-MS

First Industrial High Power Laser Perforation: Operation and Deployment

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This paper presents an overview of the advances achieved and the future ahead for the field deployment of high-powered lasers in subsurface for cased and open-hole applications. This innovation facilitates non-damaging well perforating, fracturing, descaling to improved near-wellbore communication and reduce the energy required for fracture initiation. The technology has been proven through successful lab and field demonstrations, which showcased the effectiveness of laser perforation of cased wells and penetration of various rock types, including carbonates, shales, sandstones, and more. These successes have driven the development of the pioneering and versatile high-power laser system.

Designed for safety and environmental sustainability, the enclosed system comprises a laser beam generator, nitrogen tank, coiled tubing, and opto-electrical tool to direct the beam to target. Its versatility means the same system can be used across multiple operations with interchangeable tools tailored for each application. The deployment demonstrated the performance of the latest-generation tool incorporates an advanced optical control system enabling real-time operational verification, complemented by integrated sensors providing instantaneous feedback during operation. The next step is to develop the tools needed to extend the capabilities and reach of the system, including real-time characterization and post-job valudation, in addition to the customary pre- and post-perforation logging methods. Furthermore, the advances in laser-compatible sensor arrays embedded within the tool will enable continuous monitoring of parameters like luminosity, temperature, pressure, and flow measurement, allowing the operators to evaluate and asses the tool performance throughout the operation.

Introduction

Subsurface energy extraction—encompassing mining, geothermal, and hydrocarbon production—has long relied on mechanical, hydraulic, and chemical methods to access and enhance the productivity of geological formations. While these techniques have enabled the development of vast energy resources, they are increasingly constrained by technical, environmental, and economic challenges. As global energy demand increases and environmental regulations become stricter, the limitations of conventional approaches have become more pronounced, driving the search for innovative alternatives.

Perforations and hydraulic fracturing are inherently invasive and are limited in their ability to create precise, long, or directionally controlled flow paths while often causing collateral damage to the formation, wellbore instability, and the creation of a near-wellbore "damaged zone" that impairs fluid flow and requires post-job chemical cleaning. The use of explosives for perforation can induce compaction of the minerals, reducing the rock's permeability. It can cause severe damage to the completion, thus complicating subsequent stimulation or production operations. Hydraulic fracturing, while highly effective in creating conductive pathways in low-permeability reservoirs, is associated with high water consumption, chemical use, induced seismicity, and the risk of chemical contamination of groundwater resources. These methods are bound by the state of stresses in the formation, which limits their ability to target specific zones.

As the industry seeks more sustainable, safer, and practical solutions, high-power near-infrared lasers (HPLs) have emerged as a transformative alternative to conventional stimulation methods. HPLs offer the potential to drill, perforate, tunnel, fracture, ablate, anneal, and heat-treat rocks with unprecedented precision and minimal environmental footprint. Their ability to deliver energy in a controlled, contactless manner enables new approaches to subsurface engineering, from waterless fracturing to in-situ upgrading of hydrocarbons. This document reviews the state-of-the-art HPL applications for subsurface energy extraction, with a focus on the underlying physics, laboratory and field results, and the path to commercial deployment.

High-power near-infrared lasers (HPLs), specifically continuous-wave (CW) multi-kilowatt fiber lasers operating in the 1,000–1,100 nm range, are a promising solution to many of the challenges facing subsurface energy extraction. Initially developed for industrial materials processing, these lasers offer a unique combination of high power, excellent beam quality, efficiency, and the capability to deliver their energy over long distances via optical fibers. Their application to subsurface engineering is a paradigm shift, enabling a suite of new processes that are fundamentally different from traditional mechanical, hydraulic, or chemical methods.

HPLs can be used to perforate, tunnel, ablate, anneal, melt, retort, and calcine rocks with unprecedented precision and control. By delivering energy directly to the target zone, HPLs can create clean, long, and directionally controlled flow paths, enhance permeability, and modify the physical and chemical properties of the formation. The process is inherently contactless and chemical-free, reducing the risk of formation damage, contamination, and environmental impact. HPLs can operate in any orientation, independent of the in-situ stress state, and can be integrated with advanced sensing and analytics for real-time process control.

The environmental benefits of HPL technology are significant. By eliminating the need for water, chemicals, and explosives, HPLs can reduce the operational footprint, minimize waste, and lower greenhouse gas emissions. The ability to create artificial flow paths and enhance permeability without hydraulic fracturing or chemical stimulation addresses key regulatory and social concerns. In geothermal and mining applications, HPLs offer new possibilities for accessing hard-to-reach resources and improving the efficiency of energy extraction.

The versatility of HPLs arises from their ability to induce a controllable range of physical and chemical processes in rocks, depending on the laser parameters and rock properties. Table 1 summarizes the main applications of HPLs in subsurface energy extraction, highlighting their unique advantages.

Table 2 summarizes the benefits of HPL stimulation compared to other fracturing and heat-treating techniques, e.g., shaped charges, hydraulic fracturing, plasma, and microwave.

Table 1—HPL applications and unique advantages [from: (Batarseh et al., 2024, 2019, 2012; Parker et al., 2003; San Roman Alerigi et al., 2022; Seo et al., 2022)]

Application	HPL Role	Unique Advantages
Hydraulic fracturing replacement	• Create deep, wide, and clean tunnels in any given direction. • Reduce breakdown pressure	Waterless Directional and depth control, stress-independent increased permeability, Small footprint Safe Single-tool multiple functionalities Viable at high temperature/pressure
Perforation and fracture initiation	Perforate past the damaged zone in any desired direction with high control of azimuth and penetration depth.	
Permeability enhancement	Thermally driven micro-cracking, pyrolysis, decarbonation, and clay dehydration to improve inter-pore connectivity.	
Casing cutting	• Cut out windows and slots (up to 360 degrees) for various drilling and workover applications. • Pierce and create holes up to a given casing without damaging strings underneath.	
Geothermal	Tunnel or perforate hard rock to enhance permeability.	
Heavy oil/thermal	In situ steam, channelizing sands, and clay collapse	Surface infrastructure reduction
Sensing/monitoring	Real-time pyrometry, LIDAR, LIBS	Real-time feedback and imaging. Identification of sweet spots and areas of interest.

Table 2—Comparison of different stimulation techniques for perforating, fracturing, and heat-treating [from (Alrashed et al., 2019; Batarseh et al., 2024, 2021; MacGregor and Turnbull, 2007; Othman and Soliman, 2017; Reible et al., 2016)].

Technology	Mechanism	Precision & control	Water or chemical use	Formation damage	Casing damage	Commercial maturity
Shaped charges	Explosive/ mechanical	Low: only shot angular orientation and phasing. Propagation depends on maximum horizontal stress.	None	High	High	Mature
Hydraulic frac	Mechanical	Low: Propagation depends on maximum horizontal stress	Yes	High	High	Mature
Plasma	Electromagnetic/ Thermal	Moderate: plasma heat location but lacks in situ targeting	None	Low	Moderate	Early field demonstrations
Microwave	Electromagnetic/ Thermal	High: It is possible to direct energy to a preset target.	Moderate: may require enablers to enhance absorption in certain formations.	Positive: Increases permeability	None	Lab and early field trials for heating
HPL	Electromagnetic/ Thermal	High: It is possible to direct the energy to a preset target.	None	Positive: Increases permeability	None	Early field trials.

Fundamentals of laser-rock interaction

The interaction of multi-kilowatt lasers with rocks is governed by a complex dynamical system involving electromagnetic (EM), thermal, and mechanical processes, which are influenced by the laser's characteristics (e.g., wavelength, spot size, temporal modulation) and the rocks' compositional makeup (minerals, cement, organic matter, and fluids), structural arrangements (e.g., porosity, permeability, fractures, etc.), and effective physical properties (e.g., electric permittivity, thermal conductivity, heat capacity, etc.). The interplay of these mechanisms is a nonlinear and heterogeneous process that dictates how effectively energy is absorbed and subsequently transported through the material. Figure 1 summarizes the physics involved in the interaction, along with the roles of various parameters and effects.

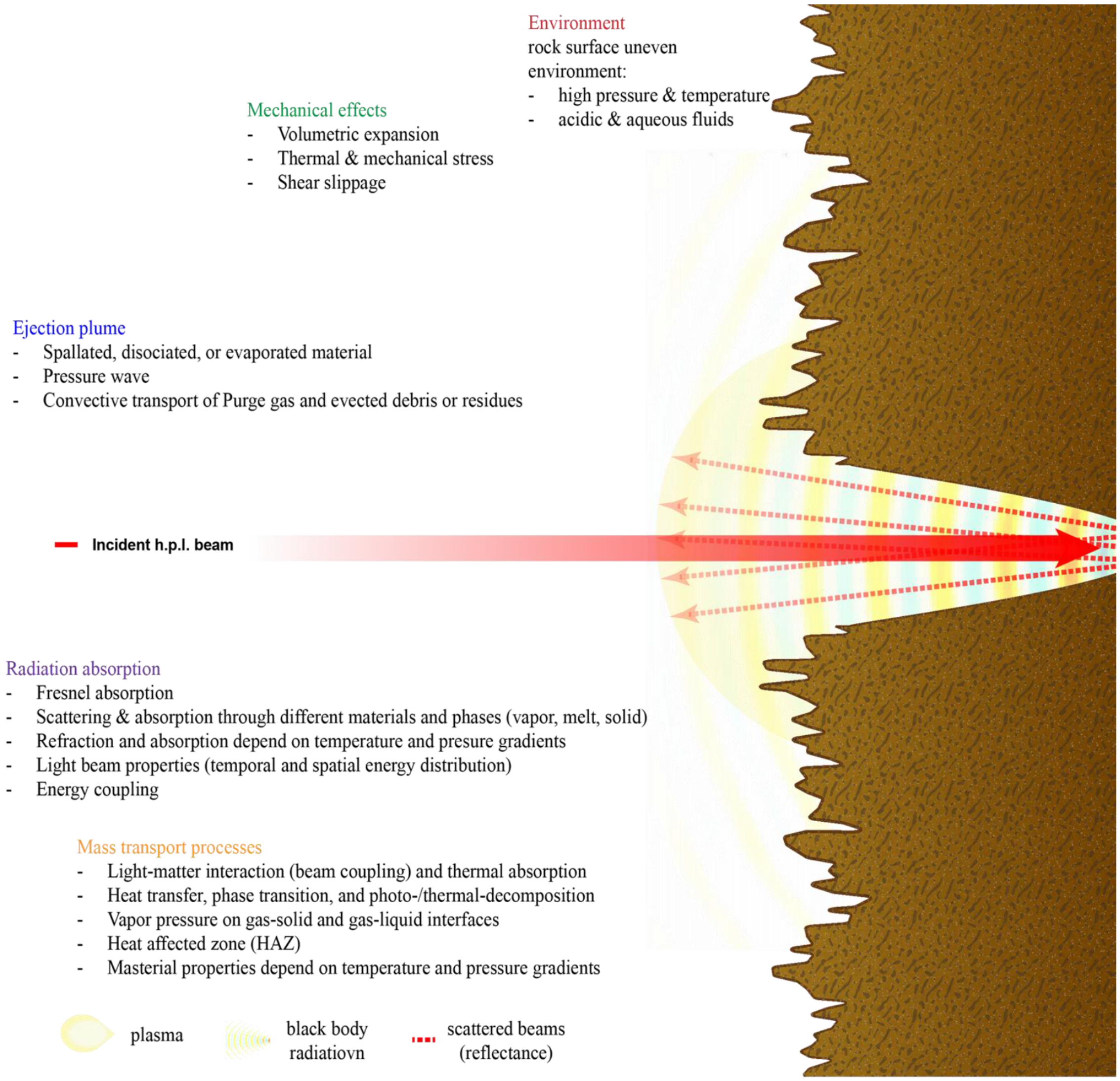

Figure 1—Fundamental physics and processes in laser-rock interaction. [source: (Batarseh et al., 2024)]

pore fluids (including water, oil, and gas). Each constituent influences the laser-rock interaction in distinct ways. Minerals such as Quartz and Feldspar (silicates) have high melting points and moderate NIR absorption, forming glassy phases upon melting. Carbonates (calcite, dolomite) have lower melting points and undergo calcining between 750 °C and 900 °C, increasing porosity and releasing CO_2. Clays (such as illite and smectite) contain bound water, which vaporizes explosively under rapid heating, promoting spallation and microfracturing. Heavy oxides (magnetite, pyrite) are strong NIR absorbers, acting as local "hot spots" for crack initiation (Erfan et al., 2018; Seo et al., 2022). Pore fluids also play a crucial role in the process, as they absorb NIR energy (especially water), vaporize rapidly, and can drive micro-explosive events that help clear debris and enhance permeability. Organic matter, such as kerogen and bitumen, exhibits strong broadband absorption in the near-infrared (NIR) range (800–2500 nm). This feature makes OM-rich zones highly responsive to HPL heating, resulting in a rapid local temperature rise, pyrolysis, and gas generation.

Rocks are inherently heterogeneous, comprising a mixture of minerals (such as quartz, feldspar, and calcite), cement, and

The energy transfer between the EM field and the rock, resulting in heating, occurs through various mechanisms due to the unique structural and compositional characteristics of the minerals involved, including in the NIR (Imanian et al., 2019; Li et al., 2022; Wei et al., 2024):

- Vibrational modes ensue from the coupling between the EM field and the molecular vibrations within the rock's mineral components. This interaction involves stretching and bending modes that correspond to the energy of near-infrared (NIR) photons. This mechanism is significant in minerals such as carbonates and water, where the vibrational overtones and combination bands of C-O and O-H bonds correspond to energy levels that can be excited by near-infrared (NIR) radiation, leading to localized heating.
- Electronic transitions occur when electrons are excited to higher energy levels upon the absorption of one or more photons. The excited electrons may undergo non-radiative decay or inelastic interactions with other particles, resulting in quantized vibrations (phonons) of the constituent molecules and lattice, which leads to localized heating. This mechanism is significant in minerals with transition metals (e.g., hematite). Multi-photon absorption is particularly relevant in high-intensity laser applications, where the energy from two or more photons can be combined to excite electrons to higher energy states, contributing to enhanced thermal effects.
- Photon-phonon interactions occur when the EM field interacts with the crystal lattice. These exciting phonons propagate through the lattice, leading to thermal effects that contribute to the overall heating of the material

The absorbed electromagnetic energy is rapidly converted into heat, raising the local temperature and triggering a cascade of physical and chemical transformations within the exposed area and the surrounding heat-affected zone (HAZ). The thermal gradient between the exposed and bulk volumes of the rock results in the development of thermal stresses, which, when exceeding the local tensile strength, lead to microcracking and spallation, especially at interfaces between minerals or fluid inclusions. Heating can also drive a series of thermo-chemical transformations ranging from pyrolysis to dehydration of clays and water-filled pores. Together, these physicochemical transformations of the material alter the optical, thermal, and mechanical properties of the rock, thereby affecting laser absorption, thermal transport, and the distribution of mechanical stresses.

In summary, the coupling between these variables is highly nonlinear, with feedback between heating, phase change, chemical transformation, and mechanical failure. The spatial and temporal evolution of temperature, stress, and chemical composition is governed by the interplay of laser parameters and rock properties. Importantly, the effective properties of the rocks emerge from the properties and distribution of their constituents, which are interdependent and vary in response to environmental and structural factors. Given this complexity, the only means of controlling the process enabling a wide range of applications, from annealing to ablation and tunneling, is through laser parameters, such as power density, wavelength, spot size, and exposure time (Batarseh et al., 2024; Erfan et al., 2018, p. 201; Han et al., 2019; Seo et al., 2022).

Optical absorption in the near-infrared range

The NIR absorption properties of rocks vary widely depending on mineralogy, porosity, fluid content, and the presence of organic matter. Table 3 summarizes typical absorption coefficients at 1064 nm for common rock types and constituents. In sedimentary rocks such as sandstones and carbonates, optical absorption can vary significantly due to differences in mineral composition. Carbonates exhibit higher absorption at longer wavelengths, which correspond to the C-O and O-H stretching vibrations (Imanian et al., 2019). Sandstones, on the other hand, show a complex absorption profile influenced by their mineralogical composition, primarily silica and various clay minerals, which result in overtone and combination bands of the Si-O vibrations (Wei et al., 2024). In igneous rocks, optical absorption is steeper in the infrared region. Basalts, rich in iron and magnesium silicates, typically display stronger absorptions at 980 nm and 1064 nm due to the presence of ferromagnesian minerals (Alshkane and Saeed, 2018; Hiryanto et al., 2023). Water and kerogens exhibit complex absorption behaviors that are significantly influenced by temperature and molecular vibrational modes. These vibrational modes correspond to the stretching and bending vibrations of O-H and C-H bonds in water and organic matter, respectively, which lead to substantial energy absorption (Batarseh, 2001; Graves et al., 1998; Othman et al., 2019; Seo et al., 2022). Importantly, the absorption profile changes significantly due to the presence of different dopants or contaminants, mineralogical diversity, temperature, and pressure.

Table 3—Optical absorption ranges for typical minerals. The provided values are indicative ranges that reflect the complexity of mineral absorption characteristics, which can vary based on factors such as composition, environmental conditions (including temperature and pressure), and measurement techniques.

Material	400 nm (cm^{-1})	532 nm (cm^{-1})	1000 nm (cm^{-1})	1100 nm (cm^{-1})	1500 nm (cm^{-1})
Quartz	0.01 – 0.02	0.01 – 0.03	0.01 – 0.03	0.02 – 0.05	0.03 – 0.1
Feldspar	0.02 – 0.05	0.05 – 0.15	0.05 – 0.15	0.1 – 0.2	0.15 – 0.3
Plagioclase	0.02 – 0.05	0.05 – 0.15	0.05 – 0.15	0.1 – 0.2	0.15 – 0.3
Calcite	0.1 – 0.2	0.15 – 0.25	0.2 – 0.3	0.3 – 0.4	0.4 – 0.5
Dolomite	0.1 – 0.2	0.15 – 0.3	0.2 – 0.3	0.3 – 0.45	0.4 – 0.5
Hematite	0.5 – 1.3	0.7 – 1.5	0.5 – 1.5	0.8 – 1.8	1.0 – 2.0
Kaolinite	0.02 – 0.05	0.03 – 0.1	0.03 – 0.1	0.05 – 0.15	0.1 – 0.3
Illite	0.03 – 0.06	0.05 – 0.15	0.05 – 0.15	0.1 – 0.15	0.2 – 0.4
Kerogens	0.1 – 0.3	0.15 – 0.35	0.2 – 0.5	0.3 – 0.6	0.4 – 0.7

Laboratory experiments and results

Laboratory investigations of HPL-rock interaction have been conducted since the late 1990s, with pioneering work by Batarseh, Graves, and others at the Colorado School of Mines (Batarseh, 2001; Graves et al., 1998). Over the next two decades, the studies primarily employed high-power fiber lasers with output powers of up to 20 kW and a wavelength of approximately 1064 nm to irradiate rock samples under controlled conditions. Given the complex interplay described earlier, these analyses employed advanced characterization methods such as thin-section petrography, X-ray diffraction (XRD) and fluorescence (XRF), scanning electron microscopy (SEM), UV-VIS-NIR reflectance spectrometry, Fourier-transform infrared (FTIR) spectroscopy, near-surface permeability injection, and various mechanical tests to map the compositional and microstructural transformations induced by laser-rock interaction, and thus devise potential applications ranging from laser perforating to multiphysics subsurface characterization (see works by Batarseh et al., 2024, 2022, 2019; San Roman Alerigi et al., 2023, 2022; Seo et al., 2022).

Traditionally, perforating and breaking rocks require high pressure to overcome the mechanical failure of the formation. When using an HPL, the electromagnetic energy absorbed by the rock causes localized heating through the process described earlier, leading to thermal stress, decomposition, and dissociation within the heat-affected zone (HAZ) of the rock. This process can be precisely controlled to achieve either application. Table 4 summarizes the distinct features of HPL rock processing for stimulation.

Table 4—Summary of key features and ranges for HPL applications

Feature	Value/range
Tunnel diameter	5 - 100 mm
Tunnel length	> 1 m

Feature	Value/range
Perforation speed	> 10 mm/s
Porosity	2.5 - 10 %
Permeability	10 – 1000 %
Break down pressure reduction	> 50%

Laser perforating

HPLs have been shown to perforate all types of rocks, including hard igneous rocks (such as granite and basalt) and soft sedimentary rocks (such as sandstone, shale, and carbonate). The process is independent of consolidation, stress state, and heterogeneity/lamination; Figures 2 and 3 present examples of laser-perforated carbonates and sandstones in different settings. In carbonates, laser heating drives a calcining process that generates a jet of CaO and CO_2, creating a hole encased with a highly permeable CaO layer (Othman et al., 2019). In sandstones, the heating process collapses clays, weakens cementations, and causes grain expansion that leads to the spallation of silica grains, also resulting in a smooth hole. This process can be precisely controlled to create holes in any direction (see Figure 3) and induce fractures while improving the rocks' permeability by several orders of magnitude (Batarseh, 2001; Seo et al., 2022). The increase in permeability can be attributed to microcracking and spallation induced. This laser-driven stimulation facilitates fluid flow in geothermal and hydrocarbon reservoirs.

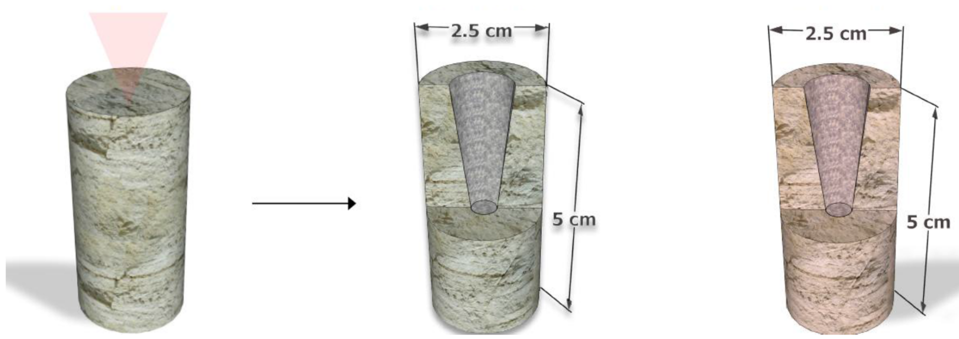

Figure 2—Micro-CT imaging of one pre- and two post-lasered samples: Indiana limestone before and after (left and center) and desert pink limestone after (right).

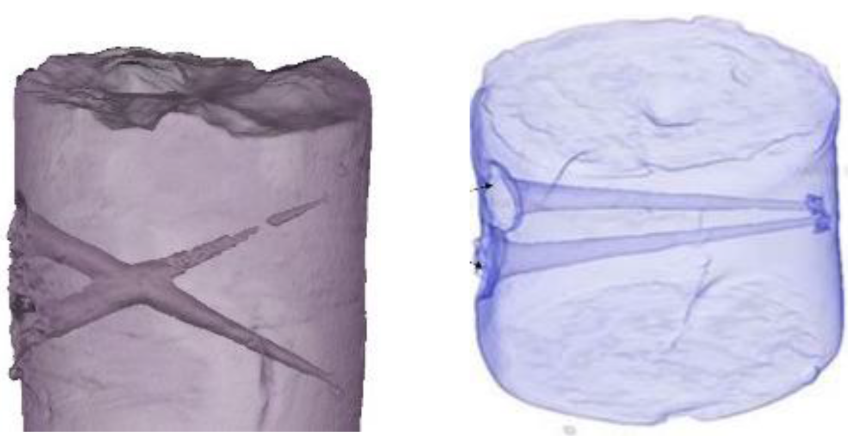

Figure 3—CT scans of laser-perforated tunnels in core samples, a carbonate (left) and unconsolidated sandstone (right). The laser process can create tunnels in any direction, regardless of the state of stress, thus enabling the creation of complex flow networks.

Fracturing

The laser-driven process can anneal or weaken rocks, thereby weakening and fracturing them. Figure 4 shows the potential fracturing effects of HPL. The results of these tests reveal that lasers can indeed initiate fractures in all rock types, including granite. In an experiment, an HPL was used to reduce the stress of a limestone sample. Prior to laser treatment, the breakdown pressure of the sample was measured at 12,150 psi, and it was then reduced to 3,800 psi after the laser treatment, a 68% reduction. This effect has been further corroborated in other experiments for fracturing (S. I. Batarseh et al., 2023; Seo et al., 2022)

Figure 4—High-power laser fracturing initiation from granite to sandstones (Batarseh, 2001).

Role of organic matter

Organic matter (OM) plays a critical role in HPL-driven processes. OM, including the thermal decomposition (retorting) of OM, occurs at temperatures ranging from 350 °C to 550°C, producing oil, gas, and char. The effects include permeability improvement, enhanced porosity, and changes in the molecular distribution of the organic content. The results of the analyses suggest that the laser can drive forward the maturity of the source rock. The expansion of gases from the breakdown of organic matter (OM) can induce microfracturing and enhance permeability. OM also lowers the effective thermal conductivity and mechanical strength of the rock, making it easier to tunnel or perforate with the laser or fracture with high-pressure liquids. These results could be critical for the application of lasers in organic-rich shales (ORS), which are typically found in unconventional reservoirs. (Sameeh I. Batarseh et al., 2022a; San Roman Alerigi et al., 2021).

HPL presents numerous advantages compared to shaped charge perforating and hydraulic fracturing across different metrics, as summarized in Table 5.

Table 5—Comparison of laser (HPL), shaped charge, and fracturing techniques for physical stimulation of rocks (Abobaker et al., 2021; Almarri et al., 2022; Batarseh et al., 2024, 2021; Han et al., 2019; Reible et al., 2016).

Metric	HPL	Shaped charge	Fracturing
Channel quality and directionality	Creates long (>1m) and directional tunnels in any formation type, regardless of stress state, with enhanced permeability	Produces irregular channels with varying penetration depth and is subject to the formation's state of stresses.	
Formation damage	Improved permeability by enhancing pores and pore interconnectivity without compacting grains	Reduce permeability due to grain compaction and chemical degradation (fracturing).	
Clean up	Not required. It produces tunnels with enhanced permeability.	Requires acidizing to clean up and remove damaged areas	Includes acids to clean up and remove damaged areas
Stress-dependence	The tunnel's direction is independent of the formation's state of stress.	Direction and propagation distance function of the state of stresses of the formation.	
Environmental impact	Chemical and water-free methods.	It has a limited environmental impact; however, its effects on the	Consumes high amounts of water and chemicals.

Metric	HPL	Shaped charge	Fracturing
	A versatile system that can cut, perforate, fracture, descale, and heat treat on a single trip. HPL source can operate with renewable energy, if available.	formation may require cleanup and stimulation.	. Chemical and water injection and disposal may have adverse environmental effects.
Safety requirements	Self-contained with multiple interlocks to prevent misfires. . Requires operation under the supervision of a safety laser officer. It does not require special permissions for transport.	In some regions, police escorts are required during transport and operation.	N/A

Lab to the field: Field trials and real-world applications of HPL

Field trials of HPL technology for decommissioning, descaling, and perforating have demonstrated its feasibility, versatility, and advantages in real-world environments. The outcome of these tests is summarized in Table 6. Importantly, all these techniques can be achieved using the same laser source and with minor modifications to the optical assembly. The latter is the optomechanical and optical system that steers and controls the beam.

Table 6—High-level outcomes of field tests of high-power laser technology in real environments.

Application	Field test	Observations	Business impact
Decommissioning (ARPA-E, n.d.)	Controllable to full-cut or melt-through of metallic producing and production infrastructure at high speeds in harsh environments.	HPL can cut and pierce at high speed with controlled angular motion through multiple (quasi-)concentric casings, including cement, in surface and aquatic environments.	Reduce costs and time by up to 50% through decommissioning infrastructure and wells (P&A) in onshore and offshore environments.
Descaling (Sameeh I Batarseh et al., 2022; San Roman Alerigi et al., 2022)	Scale removal in surface flowlines, including sections with ankylosed and saturated calcium carbonate scale.	HPL entirely removed the scale without damaging the substrate at a rate of 0.1 to 1 foot per minute, depending on the scale's hardness and saturation. . The speed of the process could be increased by improving the conveyance and scale removal mechanisms.	Reduce waste while enabling the reusing and recycling of metallic pipes. . Minimizes downtime and extends the lifetime of the producing infrastructure. . Improve flow through producing infrastructure.
Perforating (S. Batarseh et al., 2023)	Perforating completion and formation (carbonate and shale) below 600 ft in water-filled wells.	HPL produces clean pierces and slotted cuts through completion and formation, enhancing injectivity without damaging the completion or compromising wellbore integrity. . The reach and efficiency of the tool will be significantly enhanced in future generations.	Create perforations networks unrestricted by the formation's state of stress. Improve injectivity by enhancing porosity and permeability. Preserve wellbore integrity during perforating, tunneling, and fracturing operations.

Challenges, opportunities, and future work

The current focus is expanding the tool's reach through the development of high-efficiency and compact high-power laser sources in the near-infrared range, as well as optical fibers and cables for optical power delivery and data streaming, passively cooled optics, high-temperature electronics, real-time laser-based sensors, and slimmer optomechanical and optical assemblies. Table 7 provides a high-level summary of these challenges. The successful development and integration of these components present unique challenges whose solutions would enable the widespread deployment of the technology and could resonate far beyond energy extraction.

Table 7—Current challenges in the development of deployable high-power laser tools (Batarseh et al., 2024).

Component	Challenge	Remarks
High-power laser source	Develop high-efficiency (>50%) and low-cost solid-state or semiconductor lasers with output power above 10kW in the visible (400 nm and 532 nm) and near-infrared (1300 nm – 1600 nm) range of the electromagnetic spectrum.	Commercial HPLs are typically based on Ytterbium and operate at 1064 nm. Longer wavelengths enable longer transmission distances via silica optical fibers.
Optical fibers and cables for optical power delivery in the subsurface.	Transmitting multi-kilowatt NIR laser power over kilometers using fiber requires the use of materials with low loss (<10%/km). The fiber must be encapsulated in a cable for conveyance to insulate it from the environment and reduce mechanical stress.	Recent demonstrations have achieved 5 km transmission at a loss of 0.7 dB/km and 200 m at 1 dB/km in the lab and the field, respectively. Cabling remains a key challenge.
Bidirectional data transmission through optical power delivery fiber.	Enable bidirectional, low-latency, and high-bandwidth data transmission using the optical power delivery fiber concomitant to HPL transmission.	The realization of this component could significantly reduce the complexity of the cable. It would also enable real-time imaging, sensing, and control.
Real-time and HPL-compatible sensors and analytics.	Enable real-time sensing and analysis of the tool components' integrity, laser-material coupling, laser process performance, and formation and environment properties.	Real-time sensing is essential for process control and optimization, enabling operators to "see ahead of the beam" and detect anomalies. Lab and field experiments have demonstrated the viability of luminosity and spectrometry, respectively, for real-time process guidance and material identification.
High-temperature optical elements	Passively cooled and highly conductive optics with high-performance reflection and transmission, as needed.	Conventional optical assemblies for high-power lasers rely on water cooling to ensure precise beam control. Subsurface optical elements must be able to dissipate heat rapidly with minimal thermal distortion.

Ongoing work is resolving these challenges while pursuing the stepwise demonstration of components and applications. The next significant milestone is the conveyance of optical power up to one kilometer, as well as the performance of slotted cuts and perforations in both cased and open-hole conditions.

Conclusions

High-power near-infrared lasers represent a paradigm shift in subsurface energy extraction, offering precise, controllable, and environmentally friendly alternatives to conventional techniques. Laboratory and field studies have demonstrated the ability of HPLs to perforate, tunnel, and modify a wide range of rock types, enhancing permeability. Its field deployment will pave the way for creating new wellbore architectures with enhanced flow performance. The technology's versatility, from waterless fracturing to in-situ upgrading of hydrocarbons, addresses many of the technical and sustainability challenges facing the energy industry. While technical challenges remain, ongoing advances in fiber optics, sensing, and tool design are rapidly bringing HPL technology closer to deployment. HPLs are poised to play a transformative role in the future of mining, geothermal, and hydrocarbon production.

References

Abobaker, E., Elsanoose, A., Khan, F., Rahman, M.A., Aborig, A., Noah, K., Butt, S., 2021. A New Assessment of Perforation Skin Factor for Vertical Perforated Wells in Near-Wellbore Region. *J. Pet. Explor. Prod. Technol*. **12**, 117–133. https://doi.org/10.1007/s13202-021-01326-8

Almarri, M.J., AlTammar, M.J., Alruwaili, K.M., Fowler, G., 2022. Enhancing the Fracture Growth Uniformity of Perforation Clusters by Pre-injection of Cold Water. *ACS Omega*.

Alrashed, A., Miskimins, J., Tura, A., 2019. Optimization of hydraulic fracture spacing through the investigation of stress shadowing and reservoir lateral heterogeneity, in: Proceedings of the 2019 SPE Middle East Oil and Gas Show and Conference. OnePetro, Manama, Bahrain, p. SPE-195071-MS. https://doi.org/10.2118/195071-MS

Alshkane, Y.M., Saeed, S.H., 2018. Effect of Freezing and Thawing on Physical and Mechanical Properties of Sedimentary Rock. *J. Garmian Univ*. **5**, 126–140. https://doi.org/10.24271/garmian.332

ARPA-E, n.d. *High Power Laser Decommissioning Tool | ARPA-E [WWW Document]*. URL https://arpa-e.energy.gov/programs-and-initiatives/search-all-projects/high-power-laser-decommissioning-tool (accessed 6.20.25).

Batarseh, S., San Roman Alerigi, D., Marshall, S., Graves, R., 2023. First Industrial High-Power Laser Perforation Deployment, in: Proceedings of the SPE Annual Technical Conference and Exhibition 2023. Presented at the SPE Annual Technical Conference and Exhibition 2023, OnePetro, Houston, TX, p. SPE-215016-MS.

Batarseh, S.I., 2001. *Application of Laser Technology in the Oil and Gas Industry: An Analysis of High Power Laser-Rock Interaction and its Effect on Altering Rock Properties and Behavior (PhD Thesis)*. Colorado School of Mines.

Batarseh, S.I., Abass, H.H., Al-Mulhem, A.A., Habib, N.S., 2012. High Power Laser Application in Openhole Multiple Fracturing with an Overview of Laser Research; Past, Present and Future, in: SPE Saudi Arabia Section Technical Symposium and Exhibition. *Society of Petroleum Engineers*, Khobar, Saudi Arabia, pp. 347–356. https://doi.org/10.2118/160836-MS

Batarseh, S.I., Al Harith, A., Othman, H., Al Badairy, H., 2019. Unleash Unconventional Resources with the Power of Light-Laser Technology. *Society of Petroleum Engineers (SPE)*, Manama, Bahrain, p. SPE-195094-MS. https://doi.org/10.2118/195094-MS

Batarseh, S.I., Alerigi, D.P.S.R., Marshall, S., BuHassan, S., 2024. Validation of the First Industrial High-Power Laser Perforation Deployment. Presented at the SPE Annual Technical Conference and Exhibition, p. D031S034R006. https://doi.org/10.2118/220900-MS

Batarseh, S.I., Alerigi, D.P.S.R., Zefzafy, I., Enezi, H., 2023. Reducing Hydraulic Fracturing Breaking Pressure by Using High Power Lasers. Presented at the SPE Annual Technical Conference and Exhibition, p. D021S026R005. https://doi.org/10.2118/215162-MS

Batarseh, S.I., Mutairi, S., San Roman Alerigi, D.P., Al Harith, A., 2021. Laser Technology for Downhole Applications; Past, Present and Future, in: Proceedings of the 2021 SPE Middle East Oil & Gas Show and Conference. *Society of Petroleum Engineers (SPE)*, Manama, Bahrain, p. D031S016R006. https://doi.org/10.2118/204661-MS

Batarseh, Sameeh I, Mutairi, S., San Roman Alerigi, D.P., Assiri, W., Marshall, S., 2022. First industrial flowlines descaling field deployment utilizing high power laser technology, in: Proceedings of the 2022 SPE Annual Technical Conference and Exhibition. *Society of Petroleum Engineers (SPE)*, TX, USA.

Batarseh, Sameeh I., Mutairi, S.M., SanRoman, D.P., Al Harith, A.M., 2022a. The Efficiency of High Power Laser Perforation in Shale Formations. Presented at the *ADIPEC*, p. D031S076R003. https://doi.org/10.2118/211072-MS

Batarseh, Sameeh I., Mutairi, S.M., SanRoman, D.P., Assiri, W.J., 2022b. First Industrial High Power Laser Field Deployment: Lab to Field. Presented at the *ADIPEC*, p. D011S019R001. https://doi.org/10.2118/211497-MS

Erfan, M.R., Shahriar, K., Sharifzadeh, M., Ahmadi, M., Mousavi, M., 2018. Coupled T-H-M processes' effect on specific energy in continuous wave fiber laser rock perforation. *J. Laser Appl*. **30**, 032005. https://doi.org/10.2351/1.5018799

Graves, R.M., Engineering, S., Library, I., 1998. SPE 49259 StarWars Laser Technology, in: SPE Annual Technical Conference and Exhibition. *Society of Petroleum Engineers*, Louisiana, pp. 761–770.

Han, Y., Fang, Y., San-Roman-Alerigi, D.P., Batarseh, S.I., 2019. Thermal/Mechanical Interaction in Laser Perforation Process: Numerical-Model Buildup and Parametric Study. *SPE J*. **24**, 2097–2110. https://doi.org/10.2118/183836-PA

Hiryanto, Sujarwo, S., Tristanti, Santi, F.U., 2023. Elements of Metamorphic Rocks (Marble and Quartzite) of Bukit Cinta Gunung Gajah, Klaten. *J. Penelit. Pendidik. Ipa* **9**, 10433–10438. https://doi.org/10.29303/jppipa.v9i11.5260

Imanian, A., Tangestani, M.H., Asadi, A., 2019. Investigation of Spectral Characteristics of Carbonate Rocks – A Case Study on Posht Moleh Mount in Iran. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci*. **XLII-4/W18**, 553–557. https://doi.org/10.5194/isprs-archives-xlii-4-w18-553-2019

Li, X., Zuo, H., Yang, Z., Li, H., Sun, W., 2022. Coupling Mechanism of Dissipated Energy–Electromagnetic Radiation Energy During Deformation and Fracture of Loaded Composite Coal and Rock. *Acs Omega* **7**, 4538–4549. https://doi.org/10.1021/acsomega.1c06511

MacGregor, S.J., Turnbull, S.M., 2007. *Plasma channel drilling process*.

Othman, H., San Roman Alerigi, D.P., Batarseh, S.I., Al Obaid, O., 2019. Efficiency of High Power Laser in Carbonate Formations, in: Proceedings of the 2019 Abu Dhabi International Petroleum Exhibition & Conference. *Society of Petroleum Engineers (SPE)*, Abu Dhabi, UAE, p. SPE-197480-MS. https://doi.org/10.2118/197480-MS

Othman, H.A., Soliman, M.Y., 2017. Combining Microwave and Activated Carbon for Oilfield Applications: Thermal-Electromagnetic Analysis of Activated Carbon, in: SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition. *Society of Petroleum Engineers*. https://doi.org/10.2118/188130-MS

Parker, R., Xu, Z., Graves, R., Gahan, B., 2003. Drilling Large Diameter Holes in Rocks Using Multiple Laser Beams. *Congr. Appl. Laser 6*.

Reible, D.D., Honarparvar, S., Chen, C.-C., Illangasekare, T.H., MacDonell, M., 2016. Environmental Impacts of Hydraulic Fracturing, in: Orszulik, S. (Ed.), *Environmental Technology in the Oil Industry*. Springer International Publishing, Cham, pp. 199–219. https://doi.org/10.1007/978-3-319-24334-4_6

San Roman Alerigi, D.P., Batarseh, S.I., Assiri, W., 2021. The effect of high power laser on organic-rich shales, in: Proceedings of the 2021 SPE Middle East Oil and Gas Show and Conference. *Society of Petroleum Engineers (SPE)*, Manama, Bahrain, p. D031S026R001. https://doi.org/10.2118/204809-MS

San Roman Alerigi, D.P., Li, W., Sepulveda, A.C., Batarseh, S., 2023. AI inference of multiphysics properties using hyperspectral data, in: Optical Technology and Measurement for Industrial Applications Conference. Presented at the Optical Technology and Measurement for Industrial Applications Conference, SPIE, pp. 26–33. https://doi.org/10.1117/12.3005526

San Roman Alerigi, D.P., Mutairi, S., Batarseh, S., Assiri, W., 2022. Principles and Advantages of High-Power Lasers for Descaling Surface Equipment, in: Proceedings of the 2022 SPE Annual Technical Conference and Exhibition. OnePetro, Houston, Texas, USA. https://doi.org/10.2118/209977-MS

Seo, Y., Lee, D., Pyo, S., 2022. The interaction of high-power fiber laser irradiation with intrusive rocks. *Sci. Rep*. **12**, 680. https://doi.org/10.1038/s41598-021-04575-z

Wei, L., Wu, J., Li, Y., Wu, L., Fu, B., 2024. Simulation Study on the Thermal Effect of Continuous Laser Heating Quartz Materials. *Front. Chem*. 12. https://doi.org/10.3389/fchem.2024.1435562